03 Volcano plot

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Setup

```
if (!exists("project_root")) {
  cur <- tryCatch(knitr::current_input(), error = function(e) NULL)
  path <- if (is.null(cur)) NULL else if (is.character(cur)) cur else cur[["path"]]
  if (!is.null(path) && length(path) > 0) {
    project_root <- normalizePath(file.path(dirname(path[1]), ".."), mustWork = FALSE)
  } else {
    wd <- getwd()
    project_root <- if (dir.exists(file.path(wd, "Data"))) wd else normalizePath(file.path(wd, ".."), mu
  }
}
INPUT_DIR    <- file.path(project_root, "Results", "02_DEG")
RESULTS_DIR  <- file.path(project_root, "Results", "03_Volcano")
TUMOR_VS_NORMAL <- file.path(INPUT_DIR, "Tumor_vs_normal.csv")
```

Load packages for plotting

```
library(ggplot2)   # plotting
library(ggrepel)   # labels (geom_label_repel)
```

Load results

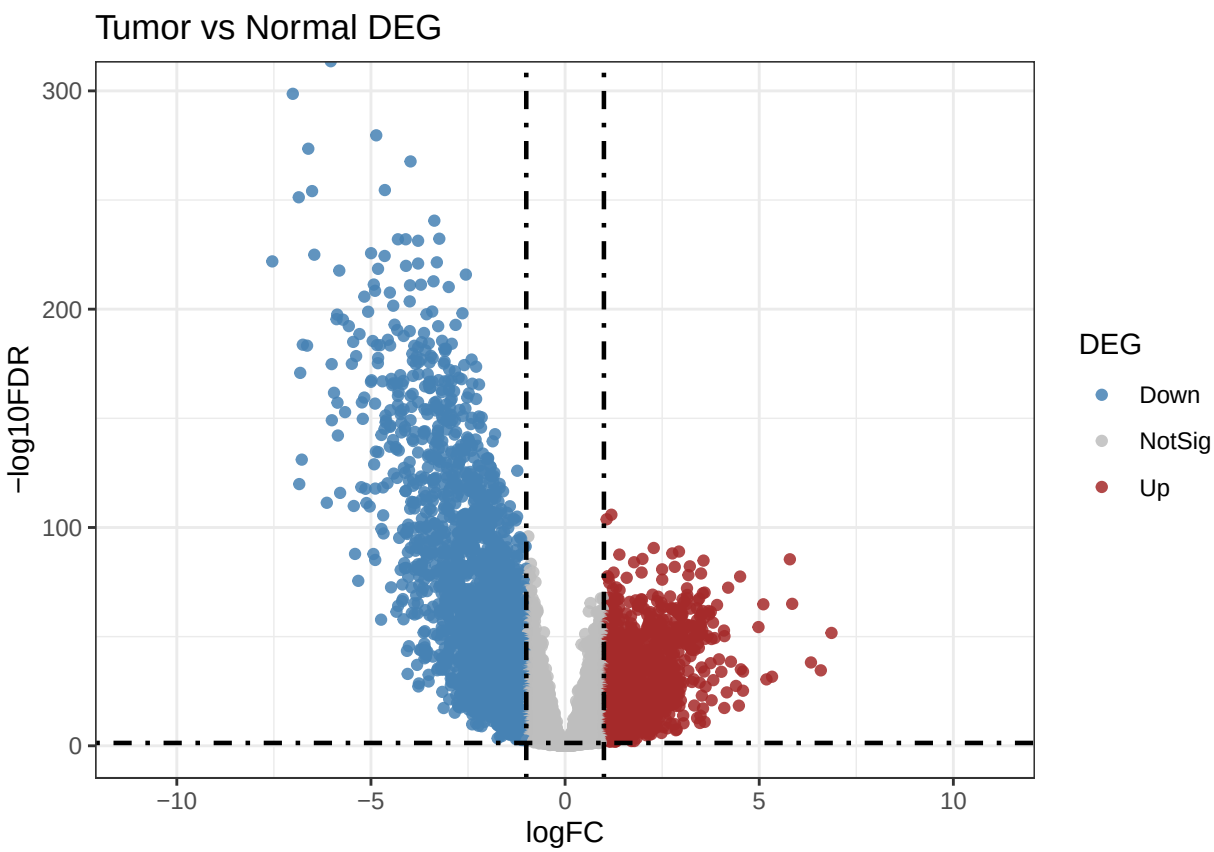
```
res <- read.csv(TUMOR_VS_NORMAL, row.names = 1)
stopifnot("logFC" %in% names(res), "adj.P.Val" %in% names(res))
res$log10FDR <- -log10(res$adj.P.Val)
res$DEG <- "NotSig"
res$DEG[res$logFC > 1 & res$adj.P.Val < 0.05] <- "Up"
```

```
res$DEG[res$logFC < -1 & res$adj.P.Val < 0.05] <- "Down"
res$DEG <- factor(res$DEG, levels = c("Down", "NotSig", "Up"))
# Label: use gene_name if present, else row name
res$gene_label <- if ("gene_name" %in% names(res)) res$gene_name else rownames(res)
message("DEGs: ", sum(res$DEG != "NotSig"), " (Up: ", sum(res$DEG == "Up"), ", Down: ", sum(res$DEG ==
```

```
## DEGs: 4334 (Up: 1552, Down: 2782)
```

Volcano plot (unlabeled)

```
dir.create(RESULTS_DIR, showWarnings = FALSE, recursive = TRUE)
xmax <- max(11, ceiling(max(abs(res$logFC))))
p1 <- ggplot(res, aes(x = logFC, y = log10FDR, colour = DEG)) +
  geom_point(alpha = 0.85, size = 1.5) +
  scale_color_manual(values = c("steelblue", "gray", "brown")) +
  xlim(c(-xmax, xmax)) +
  geom_vline(xintercept = c(-1, 1), lty = 4, col = "black", lwd = 0.8) +
  geom_hline(yintercept = -log10(0.05), lty = 4, col = "black", lwd = 0.8) +
  labs(x = "logFC", y = "-log10FDR") +
  ggtitle("Tumor vs Normal DEG") +
  theme(plot.title = element_text(hjust = 0.5), legend.position = "right", legend.title = element_blank
  theme_bw()
print(p1)
```



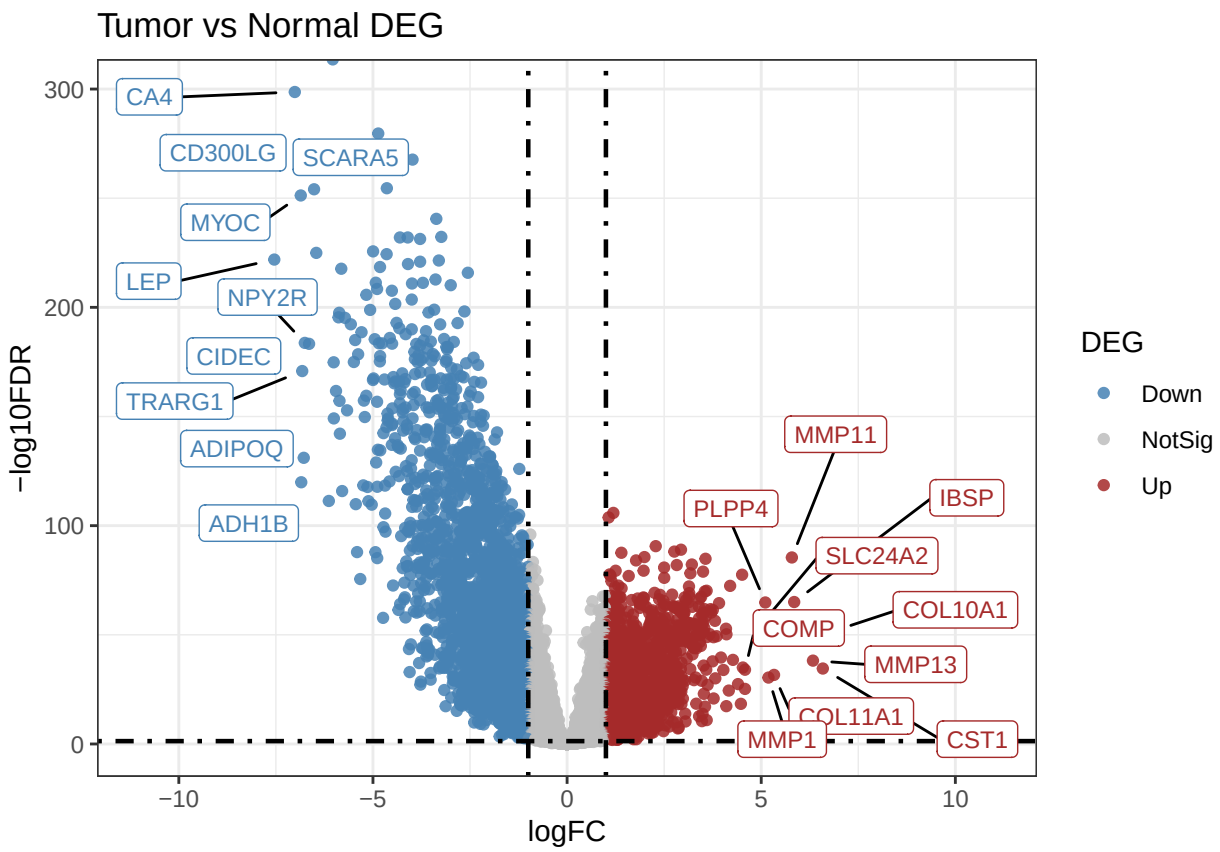

```
ggsave(file.path(RESULTS_DIR, "volcano_unlabeled.pdf"), p1, width = 7, height = 6)
```

Add labels (top 10 by logFC on each side)

```
res$label <- ""
# Top 10 Up (highest logFC)
idx_up <- order(res$logFC, decreasing = TRUE)[seq_len(min(10, sum(res$logFC > 0)))]
res$label[idx_up] <- res$gene_label[idx_up]
# Top 10 Down (lowest logFC)
idx_down <- order(res$logFC, decreasing = FALSE)[seq_len(min(10, sum(res$logFC < 0)))]
res$label[idx_down] <- res$gene_label[idx_down]
# Optional: add or overwrite specific genes by name (e.g. res$label[res$gene_label == "SPRY2"] <- "SPRY
# Optional: batch: res <- res %>% mutate(label = ifelse(gene_label %in% c("CAPN11","VEGFD","PAMR1"), ge
```

Volcano plot with labels

```
xmax <- max(11, ceiling(max(abs(res$logFC))))
p2 <- ggplot(res, aes(x = logFC, y = log10FDR, colour = DEG)) +
  geom_point(alpha = 0.85, size = 1.5) +
  scale_color_manual(values = c("steelblue", "gray", "brown")) +
  xlim(c(-xmax, xmax)) +
  geom_vline(xintercept = c(-1, 1), lty = 4, col = "black", lwd = 0.8) +
  geom_hline(yintercept = -log10(0.05), lty = 4, col = "black", lwd = 0.8) +
  labs(x = "logFC", y = "-log10FDR") +
  ggtitle("Tumor vs Normal DEG") +
  theme(plot.title = element_text(hjust = 0.5), legend.position = "right", legend.title = element_blank
  geom_label_repel(aes(label = label), size = 3, box.padding = unit(0.5, "lines"),
    point.padding = unit(0.8, "lines"), segment.color = "black", show.legend = FALSE, max.overlaps = 10
  theme_bw()
print(p2)
```




```
ggsave(file.path(RESULTS_DIR, "volcano_labeled.pdf"), p2, width = 7, height = 6)
```